

SEMINAR

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Computing Large Determinants Using Row-Reductions

1 Motivation

In our ODE course, we encounter the **Wronskian** of n functions:

$$W(y_1, \dots, y_n)(x) = \begin{vmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{vmatrix}.$$

In particular, we may be required to compute the Wronskian at a particular point $x = x_0$. This amounts to computing an $n \times n$ determinant.

2 The Computational Problem

For small matrices, one natural method for computing a determinant is **cofactor expansion**. For example,

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}),$$

where A_{ij} is obtained by deleting the i -th row and j -th column of A .

However, this approach quickly becomes cumbersome as n grows. Computing one determinant requires computing several smaller determinants, and those determinants require computing still smaller determinants.

Thus, for a large $n \times n$ matrix, we would like to find a more systematic way of computing its determinant.

3 A Different Approach

Instead of computing the determinant directly, we will first transform the matrix into a simpler form.

More precisely, we will use elementary row operations to transform a matrix

A

into an **upper triangular matrix**

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & u_{nn} \end{pmatrix}.$$

The determinant of a triangular matrix is particularly easy to compute:

$$\det(U) = u_{11}u_{22} \cdots u_{nn}.$$

Therefore, our strategy will be

$$\boxed{A \longrightarrow U \longrightarrow \det(U)}$$

where the first step consists of elementary row operations.

The main question is therefore:

How can we transform A into an upper triangular matrix while keeping track of what happens to its determinant?

This will lead us to **Gaussian elimination** and the effect of elementary row operations on determinants.

4 Geometric Interpretation of the Determinant

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{pmatrix},$$

where

$$\mathbf{v}_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} b \\ d \end{pmatrix}.$$

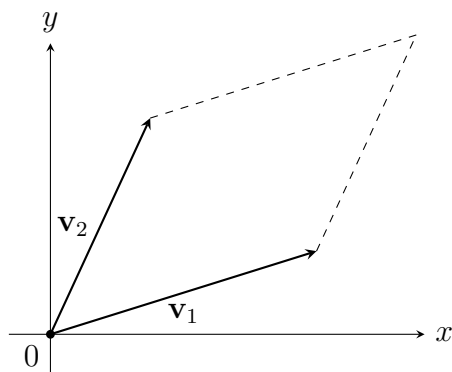
The two column vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ span a parallelogram in $\mathbb{R}^2$. Its signed area is precisely the determinant:

$$\boxed{\det(A) = ad - bc}$$

Thus,

$$\boxed{|\det(A)| = \text{area of the parallelogram spanned by the columns of } A.}$$

The sign records the orientation of the ordered pair $(\mathbf{v}_1, \mathbf{v}_2)$.



4.1 The $n \times n$ Case

The same idea extends naturally to higher dimensions.

For

$$A = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & \cdots & | \end{pmatrix},$$

the columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ span an n -dimensional parallelepiped in $\mathbb{R}^n$.

The determinant measures its signed n -dimensional volume:

$ \det(A) = n\text{-dimensional volume of the parallelepiped spanned by the columns of } A.$

Hence, we can think of the determinant as a quantity that tells us how a matrix changes volume.

5 Why $\det(AB) = \det(A)\det(B)$?

The geometric interpretation also gives us an intuition for the multiplicative property of determinants.

Consider a 2×2 matrix as a linear transformation of $\mathbb{R}^2$.

Start with the unit square. Its area is 1.

Applying B transforms the unit square into a parallelogram whose area is

$$|\det(B)|.$$

Thus,

$$1 \xrightarrow{B} |\det(B)|.$$

Now apply A to this parallelogram.

Since A scales areas by the factor $|\det(A)|$, the resulting area is

$$|\det(A)| |\det(B)|.$$

Geometrically, we therefore have

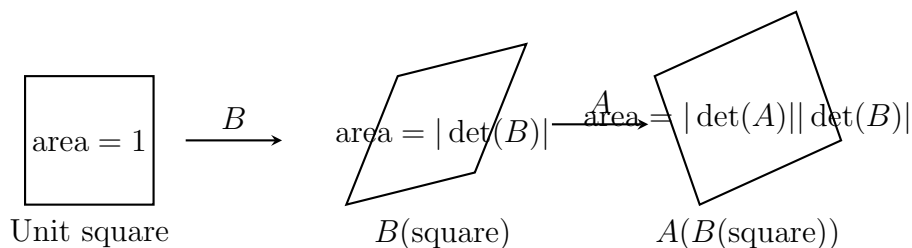
$$\boxed{1 \xrightarrow{B} |\det(B)| \xrightarrow{A} |\det(A)| |\det(B)|}$$

But applying B first and then A is exactly the composition AB . Therefore,

$$|\det(AB)| = |\det(A)| |\det(B)|.$$

When orientation is also taken into account, this becomes

$$\boxed{\det(AB) = \det(A) \det(B).}$$



The important idea is therefore:

Volume-scaling factors multiply under composition of linear transformations.

6 Elementary Row Operations as Matrix Multiplication

We now return to our original computational problem.

Our goal is to transform a matrix into upper triangular form using elementary row operations. Let us first understand one of these operations from a matrix perspective.

Consider the elementary row operation

$$R_j \longrightarrow R_j + kR_i, \quad i \neq j.$$

This operation can be represented as multiplication by a special matrix from the left.

Let I_n denote the $n \times n$ identity matrix. Define E to be the matrix obtained from I_n by replacing its (j, i) -th entry by k :

$$E = \begin{pmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ \cdots & k & \ddots & \\ & & & 1 \end{pmatrix},$$

where the k occurs in position (j, i) .

Then

$$\boxed{EA}$$

is precisely the matrix obtained from A by performing the row operation

$$R_j \longrightarrow R_j + kR_i.$$

Thus, an elementary row operation can be viewed as a **linear transformation of the rows**, implemented by multiplication with an elementary matrix.

6.1 What Happens to the Determinant?

We have already seen that

$$\det(AB) = \det(A) \det(B).$$

Therefore,

$$\det(EA) = \det(E) \det(A).$$

So everything comes down to understanding $\det(E)$.

But E is obtained from the identity matrix by adding k times one row to another:

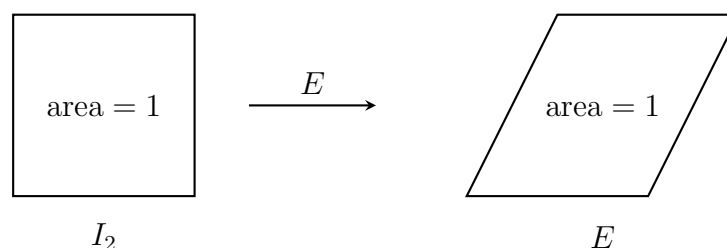
$$I_n \xrightarrow{R_j \rightarrow R_j + kR_i} E.$$

Geometrically, this transformation is a **shear**.

For example, in $\mathbb{R}^2$,

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ kx + y \end{pmatrix}.$$

This transformation slants the unit square into a parallelogram, but its area remains 1.



Hence,

$$\boxed{\det(E) = 1}.$$

Consequently,

$$\det(EA) = \det(E) \det(A) = 1 \cdot \det(A).$$

Therefore,

$$\boxed{\det(EA) = \det(A)}$$

which means that the elementary row operation

$$R_j \longrightarrow R_j + kR_i$$

does not change the determinant.

6.2 The Key Insight

We have therefore obtained an important computational tool:

Adding a multiple of one row to another does not change the determinant.

And geometrically, the reason is simple:

The corresponding linear transformation is a shear, and a shear preserves volume.

This is precisely the type of row operation we will use repeatedly to eliminate entries below the diagonal.

7 Triangularizing a Matrix

We have seen that the row operation

$$R_j \longrightarrow R_j + kR_i$$

does not change the determinant.

The natural question is now:

Can we use only this operation to transform an arbitrary matrix into an upper triangular matrix?

The answer is yes.

7.1 The Elimination Process

Consider an $n \times n$ matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

Our goal is to eliminate all the entries below the main diagonal.

Step 1: Eliminate the first column

Suppose $a_{11} \neq 0$. For every $j > 1$, choose

$$k = -\frac{a_{j1}}{a_{11}}.$$

Then perform

$$R_j \longrightarrow R_j - \frac{a_{j1}}{a_{11}}R_1.$$

The first entry of row j becomes

$$a_{j1} - \frac{a_{j1}}{a_{11}}a_{11} = 0.$$

Thus, after these operations, the matrix has the form

$$\begin{pmatrix} * & * & * & \cdots & * \\ 0 & * & * & \cdots & * \\ 0 & * & * & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & * & * & \cdots & * \end{pmatrix}.$$

The first column is now in the desired triangular form.

Step 2: Move to the second column

We now leave the first row alone and work only with the remaining rows.

Using the second row as the pivot row, we eliminate all entries below a_{22} :

$$R_j \longrightarrow R_j - \frac{a_{j2}}{a_{22}}R_2, \quad j > 2.$$

The matrix now has the form

$$\begin{pmatrix} * & * & * & \cdots & * \\ 0 & * & * & \cdots & * \\ 0 & 0 & * & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & * & \cdots & * \end{pmatrix}.$$

Step 3: Continue

We repeat the same process.

At the r -th step, use the r -th row as the pivot row and eliminate all entries below the pivot in the r -th column:

$$R_j \longrightarrow R_j - \frac{a_{jr}}{a_{rr}}R_r, \quad j > r.$$

Each step creates more zeros below the diagonal while preserving all the zeros created in the previous steps.

Eventually we arrive at

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ 0 & 0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & u_{nn} \end{pmatrix}.$$

Thus, after finitely many elementary row operations,

$$A \longrightarrow U.$$

7.2 What About a Zero Pivot?

The above description assumes that the pivot a_{rr} is nonzero.

If the pivot happens to be zero, but some entry below it in the same column is nonzero, we can first interchange the corresponding rows and continue.

For the moment, we will assume that the matrices under consideration allow us to choose a nonzero pivot at each step. The row-interchange case can be handled separately by keeping track of the change in sign of the determinant.

7.3 Putting Everything Together

Suppose the successive elementary matrices corresponding to our row operations are

$$E_1, E_2, \dots, E_m.$$

Then the sequence of row operations can be written compactly as

$$A \longrightarrow E_1 A \longrightarrow E_2 E_1 A \longrightarrow \cdots \longrightarrow E_m E_{m-1} \cdots E_1 A.$$

The final matrix is upper triangular, so

$$E_m E_{m-1} \cdots E_1 A = U.$$

Since each E_i corresponds to a shear,

$$\det(E_i) = 1.$$

Therefore,

$$\begin{aligned} \det(U) &= \det(E_m E_{m-1} \cdots E_1 A) \\ &= \det(E_m) \det(E_{m-1}) \cdots \det(E_1) \det(A) \\ &= 1 \cdot 1 \cdots 1 \cdot \det(A). \end{aligned}$$

Hence,

$$\det(U) = \det(A)$$

and therefore

$$\det(A) = \det(U) = u_{11}u_{22} \cdots u_{nn}.$$

We have therefore reduced the computation of an arbitrary determinant to the computation of the diagonal entries of an upper triangular matrix.